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Module 1.3 Firm Objectives, Costs and Revenue

1. Business Objectives

1.1 Profit Maximisation

Profit maximisation occurs where **Marginal Cost (MC) = Marginal Revenue (MR)**, provided MC is rising through MR. This is the traditional neoclassical assumption about firm behaviour — firms produce the output level where the extra cost of producing one more unit exactly equals the extra revenue earned.

At any output below MC = MR, the firm can increase profit by producing more (MR > MC, so each extra unit adds to profit). At any output above MC = MR, producing more reduces profit (MC > MR). Therefore profit is maximised precisely at MC = MR.

1.2 Revenue Maximisation

Revenue maximisation occurs where **MR = 0**. The firm produces as much as possible while each additional unit still adds to total revenue. Beyond this point, MR becomes negative and total revenue would fall.

A revenue-maximising firm produces more output and charges a lower price than a profit-maximising firm. This objective may be pursued by managers seeking to grow the business or maximise their own bonuses (which may be linked to sales revenue rather than profit).

1.3 Sales Maximisation

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Sales maximisation (Baumol) occurs where the firm sells as many units as possible while still making at least **normal profit** (i.e. AC = AR). The firm maximises output subject to the constraint of not making a loss.

This produces even more output at an even lower price than revenue maximisation. Firms may pursue this to gain market share, deter entry, or satisfy managerial growth targets.

1.4 Satisficing

Satisficing (Herbert Simon) means achieving a satisfactory level of profit rather than the absolute maximum. Firms aim for "good enough" outcomes that satisfy the demands of different stakeholder groups (shareholders, managers, workers, customers).

This is realistic because:

- Firms face uncertainty and lack perfect information about costs and revenue
- Managers may prefer a quiet life to the relentless pursuit of maximum profit
- Different stakeholders have conflicting objectives that must be balanced
- The principal-agent problem means managers (agents) may not act in shareholders' (principals') best interests

1.5 Other Objectives

- **Survival:** In recessions or during start-up phase, firms may simply aim to stay in business.
- **Growth maximisation (Marris):** Managers may seek to grow the firm as fast as possible to increase their own power, prestige, and salary.
- **Social/ethical objectives:** Social enterprises and some firms prioritise environmental or social goals alongside profit (e.g. Patagonia, The Body Shop).
- **Predatory pricing:** Temporarily pricing below cost to drive competitors out, then raising prices once dominance is achieved.

Case Study: Amazon — Growth over Profit

For most of its first two decades, Amazon prioritised revenue growth and market share over short-

term profit maximisation. Amazon reinvested virtually all revenue into expansion — new product categories, warehousing infrastructure, AWS cloud computing, and Prime membership. The company made minimal profits (and sometimes losses) for years while growing revenues exponentially. This strategy of sales/growth maximisation allowed Amazon to achieve dominant market positions in multiple sectors. By the 2020s, its market power enabled it to extract substantial profits, particularly from AWS. This illustrates how firms may rationally sacrifice short-run profit maximisation to achieve long-run dominance.

Exam tip: When discussing firm objectives, always consider the **principal-agent problem** — managers (agents) may pursue their own objectives (salary, prestige, easy life) rather than maximising profit for shareholders (principals). This is a key reason why firms may not profit-maximise in practice.

2. Costs of Production

2.1 Short Run vs Long Run

Short run: The time period in which at least one factor of production is fixed (typically capital — factories, machinery). Firms can vary labour but not their productive capacity.

Long run: The time period in which all factors of production are variable. Firms can change the scale of production (build new factories, enter/exit markets).

2.2 Fixed, Variable, and Total Costs

Cost	Definition	Behaviour	Examples
Total Fixed Cost (TFC)	Costs that do not vary with output in the short run	Constant regardless of quantity produced	Rent, insurance, salaries of permanent staff, loan repayments

Total Variable Cost (TVC)	Costs that vary directly with output	Rise as output increases	Raw materials, energy, hourly wages, packaging
Total Cost (TC)	TFC + TVC	Rises with output (starts at TFC when output is zero)	Sum of all production costs

$$TC = TFC + TVC$$

2.3 Average and Marginal Costs

$$\text{Average Fixed Cost (AFC)} = TFC / Q$$

$$\text{Average Variable Cost (AVC)} = TVC / Q$$

$$\text{Average Total Cost (ATC or AC)} = TC / Q = AFC + AVC$$

$$\text{Marginal Cost (MC)} = \text{Change in TC} / \text{Change in Q (cost of producing one more unit)}$$

2.4 Short-Run Cost Curves

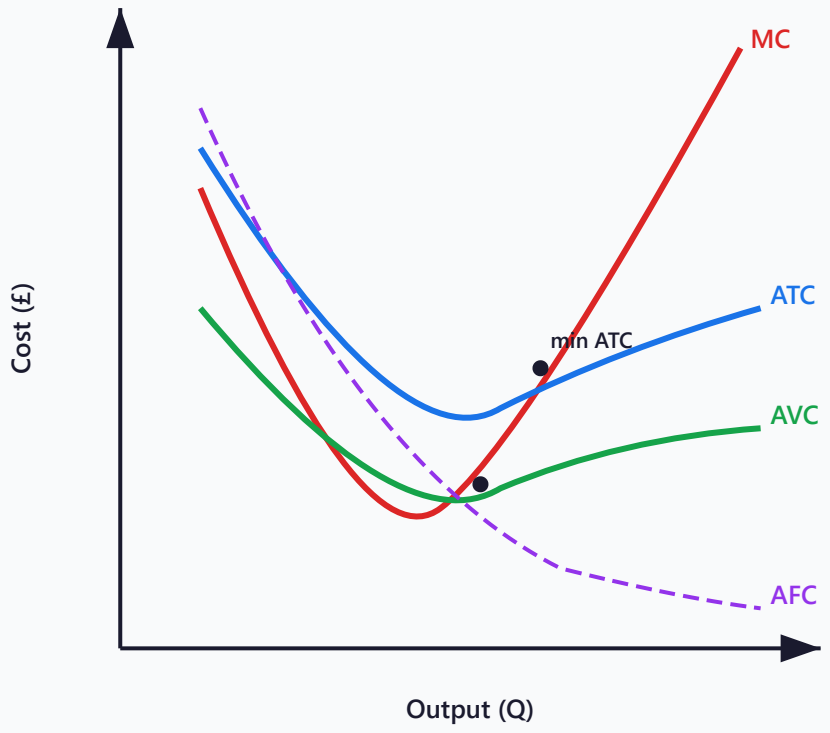

Figure 1: Short-run cost curves — MC cuts ATC and AVC at their minimum points

Key Relationships

- **MC cuts ATC at its minimum point:** When MC < ATC, average cost is falling. When MC > ATC, average cost is rising. At the intersection, ATC is at its minimum (productive efficiency).
- **MC cuts AVC at its minimum point** for the same mathematical reason.
- **AFC falls continuously** as fixed costs are spread over more units (this is why ATC and AVC converge at high output levels).
- The gap between ATC and AVC equals AFC, which narrows as output increases.

2.5 The Law of Diminishing Marginal Returns

The Law of Diminishing Marginal Returns states that in the short run, as additional units of a variable factor (e.g. labour) are added to a fixed factor (e.g. capital), there comes a point after which the marginal product of the variable factor begins to fall. This is a **short-run** concept only (at least one factor is fixed).

This law explains why the MC curve is U-shaped in the short run:

- **Initially:** As more workers are hired, specialisation and teamwork increase productivity — marginal product rises, marginal cost falls.
- **After the point of diminishing returns:** Adding more workers to the fixed capital leads to overcrowding, coordination problems — marginal product falls, marginal cost rises.

Case Study: Diminishing Returns in a Restaurant Kitchen

Consider a restaurant with a fixed kitchen space and equipment. Hiring the first few chefs increases output significantly — they can specialise (one on starters, one on mains, one on desserts). But as more chefs are added to the same kitchen space, they start getting in each other's way, waiting for equipment, and duplicating efforts. The marginal product of each additional chef falls, and the marginal cost of each additional meal rises. This is diminishing returns in action — a short-run phenomenon caused by the fixed factor (kitchen space).

3. Revenue

3.1 Revenue Concepts

$$\text{Total Revenue (TR)} = \text{Price} \times \text{Quantity} = P \times Q$$

$$\text{Average Revenue (AR)} = \text{TR} / Q = P \text{ (the demand curve is the AR curve)}$$

$$\text{Marginal Revenue (MR)} = \text{Change in TR} / \text{Change in Q (extra revenue from one more unit sold)}$$

3.2 Revenue Curves for a Price Taker (Perfect Competition)

In perfect competition, the firm is a **price taker** — it can sell any quantity at the market price. Therefore:

- AR = MR = Price (the demand curve is perfectly elastic — a horizontal line)
- TR rises at a constant rate (straight line through the origin)

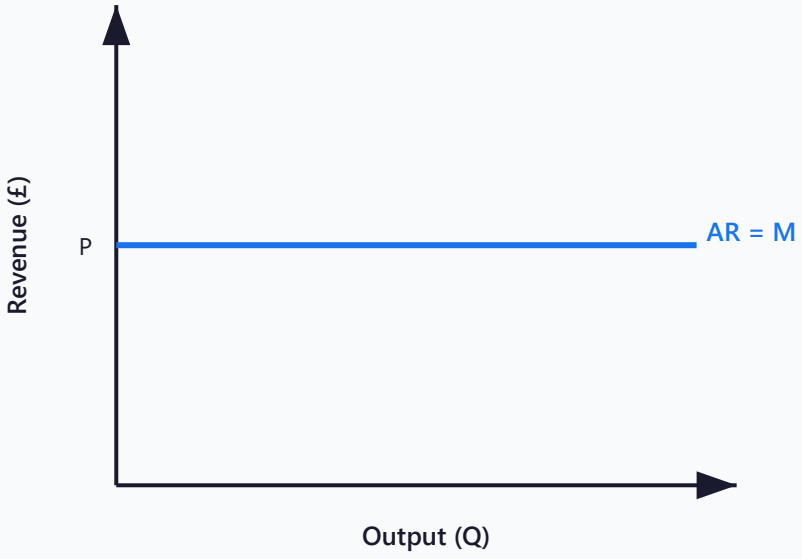

Figure 2: Revenue curves for a perfectly competitive (price-taking) firm

3.3 Revenue Curves for a Price Maker (Monopoly / Imperfect Competition)

A firm with market power faces a **downward-sloping demand curve**. To sell more, it must lower its price. Therefore:

- AR (demand curve) slopes downward
- MR lies below AR and falls twice as fast (for a linear demand curve, MR has the same intercept but twice the gradient)
- TR rises at first (while MR > 0), reaches a maximum (where MR = 0), then falls (when MR < 0)

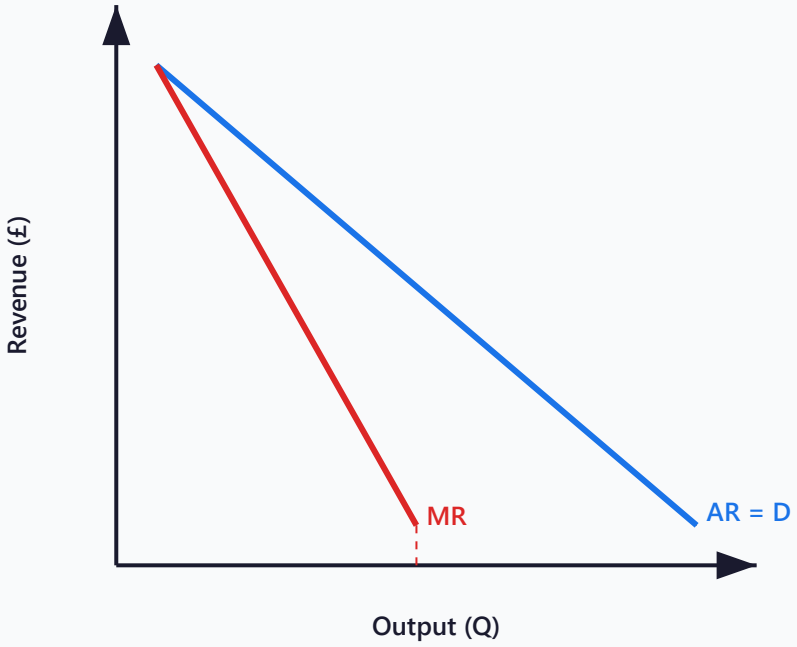

Figure 3: AR and MR curves for a price-making firm — MR falls twice as fast as AR

4. Profit

4.1 Types of Profit

$$\text{Profit} = \text{Total Revenue} - \text{Total Cost (TR} - \text{TC)}$$

$$\text{Profit per unit} = \text{AR} - \text{ATC}$$

Normal profit: The minimum return required to keep a firm in the industry in the long run. It is an economic cost (included in the cost curves). At normal profit, TR = TC (or AR = ATC).

Supernormal (abnormal) profit: Any profit above normal profit. TR > TC (or AR > ATC). This attracts new firms into the industry in the long run (unless barriers to entry exist).

Subnormal profit (loss): TR < TC. The firm is not covering all its costs. In the long run, firms making losses will exit the industry.

4.2 Profit Maximisation Diagram

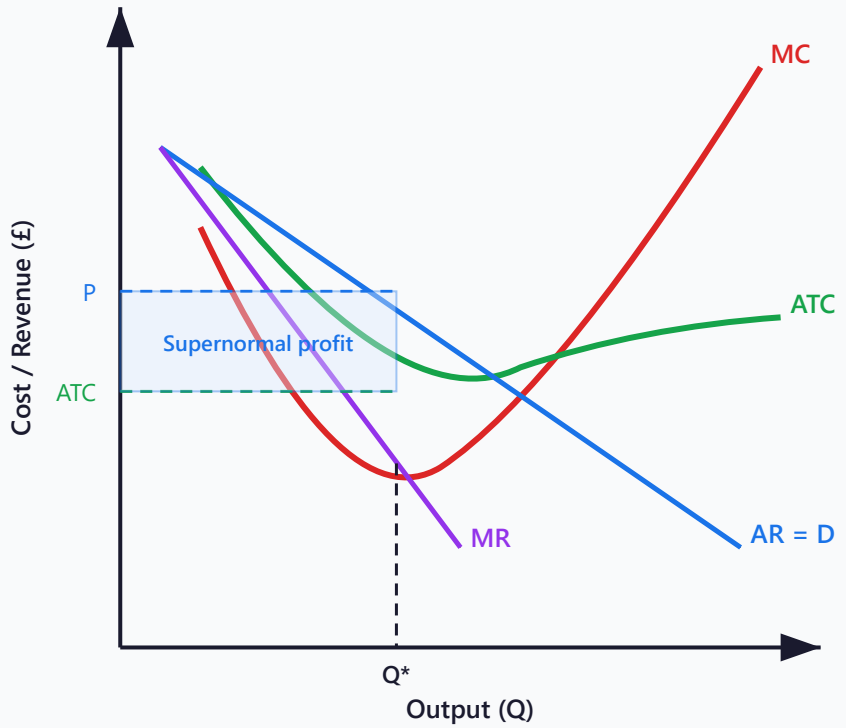

Figure 4: Profit maximisation at MC = MR, earning supernormal profit (shaded area)

4.3 The Shutdown Condition

In the **short run**, a firm should continue to operate even if making a loss, provided price (AR) covers **average variable cost (AVC)**. This is because fixed costs must be paid regardless, and any contribution above AVC helps cover them.

The firm should **shut down in the short run** if AR < AVC — it would lose more by producing than by not producing.

In the **long run**, the firm must cover all costs (AR ≥ ATC) or it will exit the industry.

5. Economies and Diseconomies of Scale

Economies of scale are the cost advantages that a firm gains as it increases its scale of production in the long run. Average costs fall as output increases. This is a **long-run** concept — all factors are variable.

5.1 Internal Economies of Scale

Cost advantages arising from the growth of the firm itself:

Type	Explanation	Example
Technical	Larger firms can use more efficient capital-intensive techniques. The cube-square law: doubling dimensions of a container increases volume eightfold but surface area only fourfold.	Oil tankers, blast furnaces, container ships
Managerial	Large firms can employ specialist managers (marketing, HR, finance) who increase efficiency.	A large firm hiring a specialist procurement manager vs a small owner doing everything
Purchasing (bulk-buying)	Large firms buy inputs in bulk at discounted prices due to bargaining power.	Tesco negotiating lower wholesale prices from suppliers than a corner shop
Financial	Large firms can borrow at lower interest rates because they are seen as lower risk by banks.	Apple issuing bonds at near-government rates vs a small firm paying high interest
Marketing	Advertising costs can be spread over a larger output, reducing the marketing cost per unit.	A national TV advert for a firm selling 10 million units vs 10,000 units

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Risk-bearing	Large diversified firms can spread risk across multiple products and markets.	Unilever selling hundreds of brands across many countries

5.2 External Economies of Scale

Cost advantages arising from the growth of the **industry** (not the individual firm):

- **Skilled labour pool:** A concentration of firms attracts specialist workers (e.g. tech workers in Silicon Valley).
- **Specialist suppliers:** Supporting industries emerge to serve the cluster (e.g. component manufacturers near car plants).
- **Knowledge spillovers:** Ideas and innovation spread between nearby firms through collaboration, staff movement, and networking.
- **Infrastructure:** Government may invest in transport and communications to support a growing industry cluster.

5.3 Diseconomies of Scale

Diseconomies of scale occur when a firm becomes so large that average costs begin to rise as output increases. The LRAC curve slopes upward beyond the minimum efficient scale.

- **Communication problems:** As firms grow, information must pass through many layers of management, leading to delays, distortion, and miscommunication.
- **Coordination difficulties:** Managing a large, complex organisation becomes harder — departments may work at cross-purposes.
- **Loss of motivation:** Workers in very large firms may feel alienated, unimportant, leading to lower productivity and higher absenteeism.
- **Bureaucracy:** Decision-making slows due to excessive procedures, paperwork, and approval chains.

Case Study: Economies of Scale at Walmart

Walmart is the world's largest retailer by revenue, and its scale generates enormous cost advantages. Purchasing economies are paramount — Walmart's buying power is so large that it can

demand the lowest wholesale prices from suppliers (some suppliers derive over 20% of their revenue from Walmart alone, giving Walmart huge bargaining leverage). Technical economies come from automated distribution centres and advanced logistics systems. Financial economies allow Walmart to borrow at very low rates. These combined economies of scale allow Walmart to operate on extremely thin profit margins (around 2-3%) while still generating substantial total profit, undercutting smaller competitors on price.

5.4 The Long-Run Average Cost (LRAC) Curve

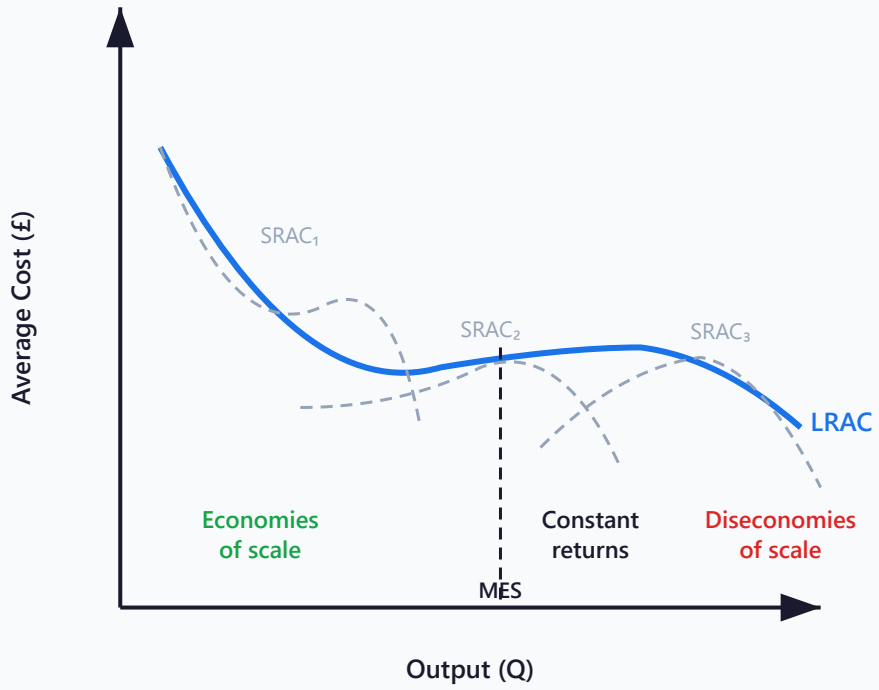

Figure 5: Long-Run Average Cost curve — an envelope of SRAC curves. MES = Minimum Efficient Scale

5.5 Minimum Efficient Scale (MES)

Minimum Efficient Scale (MES) is the lowest level of output at which all economies of scale have been exploited and LRAC is at its minimum.

The MES is significant because:

- If MES is high relative to market demand, only a few firms can operate efficiently — this creates **natural barriers to entry** and tends toward oligopoly or monopoly.
- If MES is low relative to market demand, many firms can operate efficiently — the market tends toward competition.
- Industries with high MES include: car manufacturing, aircraft production, utilities. Industries with low MES include: hairdressing, restaurants, window cleaning.

Case Study: Natural Monopoly in Water Supply

Water supply has an extremely high MES relative to the market. The fixed costs of building a pipe network are enormous, but the marginal cost of supplying an extra household is very small. It would be highly inefficient to have competing pipe networks running under every street. This makes water supply a natural monopoly — the LRAC curve is still falling over the entire range of market demand, meaning one firm can supply the whole market at lower average cost than two or more firms. This is why water companies (like Thames Water, Severn Trent) are regional monopolies regulated by Ofwat.

6. Returns to Scale (Long Run)

Returns to scale describe what happens to output when all factors of production are increased proportionally in the long run. This is distinct from diminishing marginal returns (a short-run concept).

Type	What Happens	Effect on LRAC
Increasing returns to scale	Output rises more than proportionally to the increase in inputs (e.g. doubling inputs triples output)	LRAC falls (economies of scale)
Constant returns to scale	Output rises in exact proportion to input increase	LRAC remains constant

Decreasing returns to scale	Output rises less than proportionally to input increase	LRAC rises (diseconomies of scale)

Exam tip: Do not confuse **diminishing marginal returns** (short-run, one factor fixed) with **decreasing returns to scale** (long-run, all factors variable). These are different concepts operating in different time frames.

Evaluation — Firm Behaviour in Practice

- The profit-maximisation model (MC = MR) assumes perfect information about costs and revenues — in reality, firms often lack this data and must estimate.
- Many real-world firms satisfice rather than optimise, balancing multiple stakeholder demands.
- Short-run and long-run objectives may differ — a firm may sacrifice short-run profits for long-run market share.
- The principal-agent problem means managerial objectives may diverge from shareholder interests.
- Cost structures vary enormously across industries — some have high fixed costs and low variable costs (airlines), others the reverse (consulting).

7. Deeper Evaluation: Cost Theory and Market Structure

Evaluation — Economies of Scale: Benefits vs Risks

- **Productive efficiency:** Economies of scale reduce average costs — improving productive efficiency (producing at lowest possible cost). This can benefit consumers through lower prices if competition forces firms to pass on cost savings.
- **Market power risk:** High MES relative to market demand creates natural barriers to entry, tending toward oligopoly or monopoly. The cost advantages of large firms may prevent efficient smaller firms from entering — reducing dynamic competition and innovation over time.

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- **Diseconomies in practice:** Boeing's 737 MAX crisis (2019) illustrates how diseconomies of scale in large, complex organisations can be devastating. Pressure to cut development costs (overconfidence in existing designs), poor communication between engineering and management, and insufficient oversight of safety systems contributed to two crashes killing 346 people. The organisational failures of a very large, diversified company contributed to a safety failure that a smaller, more focused organisation might have avoided.
- **X-inefficiency (Leibenstein):** Firms protected from competition by barriers to entry (including high fixed costs and MES) may not minimise costs in practice — slack and waste emerge when competitive pressure is absent. This X-inefficiency can offset the theoretical gains from economies of scale.
- **External economies — agglomeration:** Industrial clusters (Silicon Valley for tech, City of London for finance, West Midlands for automotive) generate external economies that benefit all firms, not just large ones. Policy to create or support clusters can spread benefits more widely.

Evaluation — Profit, Revenue, and the Real World

- **Pricing in practice:** Most firms use **cost-plus pricing** (also called mark-up pricing) — calculating average cost and adding a profit margin. This is simpler than MC = MR pricing and does not require precise knowledge of the demand curve. It is consistent with satisficing.
- **Limit pricing:** Incumbent firms may price below short-run profit-maximising price to deter entry — accepting lower current profit to protect future profit streams. This is a form of strategic pricing that cannot be explained by simple profit maximisation.
- **Revenue vs profit objectives in large firms:** Empirical research by William Baumol found that managerial salaries and bonuses in large US firms were more strongly correlated with firm size and revenue than with profit — supporting the revenue maximisation hypothesis for manager-controlled firms.
- **Contestable markets (Baumol):** In markets with low sunk costs and free entry/exit, even monopolies may be forced to price competitively — because potential entrants constrain pricing. This blurs the distinction between monopoly and competition in some industries (e.g., airline routes).
- **The role of uncertainty:** John Maynard Keynes noted that investment decisions are driven by "animal spirits" as much as rational calculation. Under genuine uncertainty about future demand

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and costs, firms cannot maximise profit in the neoclassical sense — they make judgement calls. This is why business confidence is such an important determinant of investment.

Case Study: EasyJet — High Fixed Costs, Low Variable Costs, and Revenue Management

EasyJet has an extreme cost structure: very high fixed costs (leasing and maintaining aircraft, airport slots, pilot contracts) and relatively low variable costs per passenger (catering, fuel is semi-variable). This means that as load factors (percentage of seats filled) rise, average costs fall rapidly — strong economies of scale from spreading fixed costs. EasyJet's response is sophisticated revenue management: dynamic pricing that rises sharply as departure approaches and seats fill up, designed to maximise revenue at each load factor. The yield management system attempts to maximise TR (not just output), reflecting a revenue-maximisation approach rather than simple profit maximisation at MC = MR. The COVID-19 pandemic devastated EasyJet — its high fixed costs meant that even with zero passengers, costs continued, rapidly exhausting cash reserves and requiring emergency equity raises.

Case Study: UK Supermarket Price Wars — Oligopoly and Pricing Strategy

The UK grocery market (Tesco, Sainsbury's, Asda, Morrisons, plus Aldi/Lidl) illustrates how oligopolistic competition shapes pricing behaviour. The "Big Four" supermarkets have historically avoided outright price competition (price wars are mutually destructive — they lower margins for all). Instead, they compete on loyalty schemes, own-brand ranges, service quality, and store format. When Aldi and Lidl began gaining market share in the 2010s-2020s (growing from under 5% to over 20% combined by 2023), the incumbents were forced to respond with price investment — Tesco's "Aldi Price Match" launched in 2020. This forced reduction in margins (Tesco's operating margin fell from ~5% to ~3%) illustrates how competitive entry can move behaviour closer to the competitive outcome, even in a concentrated market.

Exam Tips — OCR Firm Theory

- **Objectives questions:** Always consider the market structure — in perfect competition, profit maximisation is forced by competitive pressure. In monopoly, firms have discretion to satisfice or pursue other objectives. Mention this structural constraint.

- **Cost curve diagram technique:** Draw MC first (U-shaped, steep), then ATC (flatter U-shape, above MC on the left, below MC on the right). MC must cross ATC at ATC's lowest point. AVC sits below ATC, also U-shaped. AFC falls continuously (hyperbola).
- **Profit diagram:** At profit max (MC = MR), draw a vertical line to the x-axis (output Q*). Read up to AR to get price P. Read up to ATC to get average cost. The profit rectangle = (P - ATC) × Q*. Shade it and label it clearly.
- **Short run vs long run:** Diminishing returns is short-run (one factor fixed). Returns to scale is long-run (all factors variable). Do NOT confuse them — this is a common exam error that OCR examiners flag.
- **Evaluation trigger words:** "However," "on the other hand," "this depends on," "in the long run," "in practice." Use these to signal evaluation and earn the higher AO3 marks.

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